



Sangfor Cloud Product

Configuration Specification Guide

Applicable version HCI 6.11.1 & SCP 6.11.1

Document Version 1.0

2025Q2

Sangfor Technologies Inc.

A decorative graphic at the bottom of the page featuring overlapping wavy bands of green, blue, and light blue.

Revision History						
Serial number	Summary of the revisions	Revision Date	Previous revision number	Revised version number	Revised by	Approver
Note: Revision history records the basic control information currently valid when this document is submitted. The validity period of the current version of the document will automatically end when the new version of the document takes effect. When the document version number is less than 1.0, it means that the version of the document is a draft and is for reference only.						

■ Copyright Statement

Unless otherwise specified, the copyright of any text description, document format, illustrations, photos, methods, processes, etc. in this article belongs to Sangfor and is protected by relevant property rights and copyright laws. No individual or organization may copy or quote any fragment of this article in any way without the written authorization of Sangfor.

Table of content

Sangfor Cloud Product Configuration Specification Guide	1
1. Maximum configuration specifications for hosts and clusters	4
2. Maximum configuration specifications for virtual machines.....	5
3. aSAN Maximum Configuration Specifications.....	8
4. aNET Maximum Configuration Specifications.....	1 0
5. aSEC Built-in Security 2.0-Distributed IPS Configuration Specifications	1 2
6. Maximum configuration specifications for cloud management (SCP)	1 3
7. Micro-segmentation configuration specifications	1 4
8. Stretched cluster configuration specifications	1 4
9. Maximum Disaster Recovery Configuration Specifications.....	1 5
10. 2 nodes +1 witness volume configuration specifications	1 6

1. Maximum configuration specifications for hosts and clusters

CPU	Logical CPUs (CPU threads) per host	128 (for the current 6226R shipped devices, calculation: number of physical CPUs × number of cores per physical CPU × number of hyperthreads) 512 (theoretical maximum)
	NUMA nodes per host	2 (Delivery equipment) 4 (maximum)
	CPU over-provisioning ratio for each host	2x (recommended) 2x (default) Maximum (unlimited)
Memory	Physical memory per host	3TB (each model is subject to CPQ system parameters) 2 ⁶⁴ bytes (theoretical limit)
Disk	Number of physical disks per host (including SSDs and HDDs)	The maximum value is the number of disks corresponding to the server model + the number of available slots (subject to CPQ system parameters)
External Storage	The maximum number of LUNs supported by each cluster	256
	Maximum capacity of a single LUN of external storage supported	Measured maximum: 255T Theoretical maximum: 1024T Recommendation: less than 255T
	Shared disk on external storage	Cluster: supports 1000 shared disks Number of virtual machines a single shared disk can be

		mounted to: A single shared disk supports simultaneous mounting by 128 virtual machines
Network Card	Number of network cards per host (1 Gb Ethernet, 10 Gb Ethernet, 25 Gb Ethernet)	Unlimited, depends on the motherboard PCI slot
	Maximum number of network aggregation ports configured on a single host	16
Cluster	The maximum number of hosts supported by a cluster	X86: 48 units are recommended Maximum 64 units
	The number of virtual machines that can run in a cluster	Recommended within 2000 Maximum 3000 units
UPS	Maximum number of UPS supported by a single cluster	256
Recycle Bin	Default retention time interval for recycle bin files	The default is 30 days, and can be configured from 1 to 365 days

Table of maximum configuration specifications for hosts and clusters

2. Maximum configuration specifications for virtual machines

CPU	Maximum number of vCPUs in a virtual machine	1. Do not exceed the number of logical CPUs on a single host 2. The maximum measured value of Windows/Linux is 192vCPU, and the theoretical value of OS is 512 3. Maximum number of devices shipped is 96
Memory	Virtual machine memory	1. Recommended within 2TB 2. The maximum physical memory does not exceed that of a single host
Disk	Maximum number of virtual	16

	machine disks	
	Maximum capacity of a single virtual disk	Thin allocation: 63T Dynamic allocation: 16T Pre-allocated: 16T
Network Card	Maximum number of network cards that can be configured in a virtual machine	10
USB	Maximum number of USB devices that can be configured in a virtual machine	12
Graphics	Maximum number of graphics cards that can be configured for a virtual machine	8
Snapshot	Number of internal disk snapshots for a virtual machine	7
	Maximum number of external disk snapshots for a VM	32
	Maximum number of virtual machine storage snapshots	128
	Maximum number of virtual machines that a consistent snapshot group can contain	64
	Maximum number of snapshots for a consistency group	64*128 (8192)
Migration	The maximum number of VMs that can be migrated in a cluster	50
	Maximum number of concurrent VM migrations within a single host	Default: 2 Configurable: 1-10
	Maximum number of concurrent VM migrations within a single storage	Default: 1 Configurable: 1-10
	Maximum number of concurrent	1

	VM cross-cluster hot migrations	
	The maximum number of concurrent VM migrations from the managed VMware platform to HCI	10 concurrent connections per host
Backup	Backup frequency supported by the backup policy	Backup strategy: weekly backup (Monday to Sunday, 00:00 to 23:00), daily backup (00:00 to 23:00), hourly backup (1 hour, 2 hours, 3 hours, 4 hours, 6 hours, 8 hours, 12 hours) CDP strategy: 1 hour, 2 hours, 4 hours, 8 hours, 12 hours, 24 hours IO log interval: 1 second, 5 seconds
	Maximum number of concurrent backups supported	Single host normal backup concurrency: 2 single-host CDP backup: 2 Default concurrency for storage backup: 4, maximum configurable: 16 Storage backup replication default concurrency: 4, maximum configurable: 16 Default concurrency for storage archiving: 2, maximum configurable: 16 The default number of concurrent data sorting tasks for each host is 2, and the maximum configurable number is 10. Default concurrent number of storage merges: 2, maximum configurable: 16 The default number of concurrent storage backup monitoring checks is 2, and the maximum configurable number is 16.
	Maximum supported backup rate (MB/S)	Default value of backup speed limit: 80, maximum configurable: 300 The data sorting speed limit for each virtual machine is 50, and the maximum configurable speed limit is 300. Merging speed limit default value: 50,

		maximum configurable speed limit: 256, configurable unlimited speed Backup health check rate limit default value: 40, maximum configurable: 300
Clone	Maximum number of concurrent cloning tasks supported	Full clone: 2 Instant full clone: 20 Linked clones: 20
	Maximum number of clones per time	100
Power on	Number of parallel tasks	By default, each host runs 10 startup tasks in parallel. If aSAN is activated, 13 boot tasks can be run in parallel VDI virtual machines start up 30 tasks in parallel
Virtual Machine Grouping	Maximum grouping depth	5
Virtual Machine Tags	Maximum number of VM tags	4096

Maximum configuration specifications for virtual machines

3. aSAN Maximum Configuration Specifications

Storage Volumes	Maximum number of hosts supported by a single volume	Maximum value 24 Recommended value 8-12
	Minimum number of hosts supported by virtual storage	2
	Maximum number of volumes in a single cluster	Max value: 6 Recommended value 1-3 (based on actual business needs)
	Number of composite volumes supported by a single host	2 volumes
	Maximum number of storage volumes in a cluster	6 volumes

	Maximum number of disk groups per host	8
	Maximum number of disks per disk group	8
	The capacity-cache ratio of a hybrid volume must be no less than	3% recommended not less than 5%
	Maximum capacity of a single cache disk	32TB The recommended value is no more than 8TB
	Minimum capacity of a single cache disk	100G
	Maximum capacity of a single data disk	32TB The recommended value is no more than 14TB (If the recommended value is exceeded, please contact technical support to assess the impact)
	Minimum capacity of a single data disk	200G
Virtual iSCSI disk	Configurable iSCSI disks	2 host volumes or 2+1 volumes: maximum value 2048 3 host volumes: maximum value 3072 4 hosts and above: maximum value 4096
	Maximum capacity of a single virtual iSCSI disk	64T
Virtual shared disk	Maximum number of virtual shared disks that can be configured in a cluster	200
	Maximum capacity of a single virtual shared disk	4T
Virtual disk	Number of stripes	Default: Automatically adapted The number of stripes can be manually configured: 1-12

aSAN Maximum Configuration Specifications

4. aNET Maximum Configuration Specifications

Physical edge	Maximum number of configurable physical edge	1024
	Maximum number of connections for a single physical edge	2048
	Maximum number of configurable port groups per physical edge	2048
Virtual Router	Maximum number of configurable vRouters	12000
	The number of network ports that can be configured for each vRouter	511 Note: There is 1 built-in network port, which is not displayed in the topology.
	Each vRouter can be configured with the number of VLAN subnet ports	64
	The number of static routes that can be configured for each vRouter	512
	Policy routes that can be configured for each vRouter	IPv4 1024 IPv6 1024
	The number of SNATs that can be configured for each vRouter	512
	The number of DNATs that can be configured for each vRouter	512
	The number of ACLs that can be configured for each vRouter	5000
	The maximum number of DHCP4 that can be configured on a virtual router	512
	The maximum number of DHCP6 that can be configured on a virtual router	512
	Number of elastic IP addresses bound to a single shared bandwidth	127 (management plane)
Distributed	Maximum number of cluster policies	32000

Firewall	Global maximum number of rules	4096
	Maximum number of rules for a single policy	1000
	Maximum number of services in a firewall rule	10
	Maximum number of custom services	480000
	Maximum number of VM groups in a single rule	200 The number of discrete IP addresses must not exceed 2,000.
	Maximum number of IP groups in a firewall rule	200 The number of discrete IP addresses must not exceed 2,000.
	Maximum number of IP addresses in a firewall group	The number of discrete IP addresses cannot exceed 2,000.
	Maximum number of VM labels in a single rule	200 The number of discrete IP addresses must not exceed 2,000.
	Maximum number of VMs in a single rule	No restrictions The number of discrete IP addresses must not exceed 2,000.
Virtual Switch	Maximum number of configurable virtual switches	30000
	Maximum number of connections to a single distributed switch	1024
	Maximum number of network ports that can be connected to each virtual switch	1024
Traffic Mirroring Specifications	Maximum number of traffic mirroring policies supported by a single cluster	1024
	Maximum number of physical host service ports supported by the image source	64
	Maximum number of virtual network cards supported by the image source	1024

5. aSEC Built-in Security 2.0-Distributed IPS Configuration Specifications

Resource specifications for container instances	Supports maximum traffic throughput rate	vAF reference value	Remark
1vCPU 1G memory	800Mbps	50Mbps	vAF is 1C2G
2vCPU 2G memory	1.8Gbps	200Mbps	vAF is 2C4G
4vCPU 4G memory	3.3Gbps	400Mbps	vAF is 4C8G
8vCPU 8G memory	5.2Gbps	1.6Gbps	vAF is 8C16G

aSEC container instance resource specification table

Item	Maximum specifications	Remark
Intrusion Prevention Policy (HCI)	512	Refers to the entire HCI cluster (the same below)
Intrusion Prevention Policy (SCP)	256 entries/each network domain	The network effective domain refers to VPC or classic network (the same below)
Web Vulnerability Protection Policy (HCI)	512	
Web Vulnerability Protection Policy (SCP)	256 entries/each network domain	
Permanent blacklist (HCI)	2048	
Permanent Blocking Blacklist (SCP)	1024 entries/each network domain	

Rule Whitelist (HCI)	2048	
Rule Whitelist (SCP)	1024 entries/each network domain	
Cloud Host/IP Whitelist (HCI)	2048	
Cloud Host/IP Whitelist (SCP)	1024 entries/each network domain	

aSEC maximum configuration specification table

6. Maximum configuration specifications for cloud management (SCP)

Deployment Mode	Standalone mode (16C 26G)	Standalone mode (22C 26G)	Standalone mode (22C 42G)	Single Machine (30C 42G)	Cluster mode (three nodes) (32C 64G)
Maximum number of managed clusters	4	8	16	25	128
Maximum number of managed hosts	32	64	128	256	1024
Maximum number of managed virtual machines	3000	5000	10000	30000	60000
Number of	32	64	128	128	2000

tenants					
Number of users	200	500	1000	1000	4000
Concurrent Users	64	128	256	256	300

Cloud Management Maximum Configuration Specifications Table

7. Micro-segmentation configuration specifications

aNI Specifications	vCPU	Mem	Disk	Virtual machine scale	Maximum flow
Small	8	16G	500G	100	100k/5min
Medium	32	64G	1T	1000	1 million/5min
Large	64	128G	1.5T	3000	3 millions/5min

Micro-segmentation aNI Specifications

8. Stretched cluster configuration specifications

Stretched Volume	Number of fault domains	Stretched volume is fixed with 2 fault domains
	Maximum number of volumes supported	A cluster supports at most one stretched volume, but supports multiple normal volumes, a maximum of 1 stretched volume + 5 normal volumes.
	Inter-site link bandwidth	10G bare fiber, latency <1ms
	Minimum deployment specifications for witness nodes	Support virtualization deployment and physical machine deployment CPU: 8 cores and 16 threads

		Memory: 16GB Witness disk: capacity >100GB Performance >= 1000 IOPS
	Network from the witness node to both fault domains	Bandwidth>100Mbps, latency<5ms

Stretched cluster configuration specifications

9. Maximum Disaster Recovery Configuration Specifications

Disaster Recovery Policies	Maximum number of disaster recovery policies supported	256
	Maximum number of virtual machines supported by one disaster recovery policy	256
	Maximum number of local backup policies supported	No restrictions
	Maximum number of local backup VMs supported by one policy	256
	Maximum number of supported CDP policies	No restrictions
	Maximum number of CDP VMs supported by one policy	256
Business Group	Maximum number of supported business groups	1000
	Maximum number of virtual machines supported by a business group	100
Recovery Plan	Maximum number of supported recovery plans	1000
	Maximum number of business groups supported by a recovery plan	100
SCP master and backup	Number of backup SCPs supported	1
RPO	Configurable according to actual business and environment conditions	RPO = (1 second, 10 seconds, 10 minutes, 30 minutes, 1 hour, 2 hours, 4

		hours, 8 hours, 12 hours, 1 day, 2 days, one week)
RTO	Quickly restore the virtual machine startup time from disaster recovery	1-5mins

Disaster Recovery Maximum Configuration Specifications Table

10.2 nodes +1 witness volume configuration specifications

2+1 Volumes	Maximum number of volumes supported	A 2+1 cluster supports at most one volume. If you want to create another volume, you must first expand cluster to 3 hosts or more.
	Minimum deployment specifications for witness nodes	Support virtualization deployment and physical machine deployment (or witness appliance) CPU: 6 cores 12 threads Memory: 16GB Disk capacity: capacity > 64G Performance >= 1000 IOPS

2+1 Volume configuration specification table

